Deep Networks with Stochastic Depth

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CS5482 final project — Replication Study

paper

Introduction

Motivation: Deep residual networks (100+ layers) achieve higher expressiveness but suffer from:

1) vanishing gradients
2) diminishing feature reuse
3) long training times

Dilemma: Train short networks (efficient) but test deep (expressive)

Solution: Stochastic Depth

Randomly drop entire residual blocks for each mini-batch during training, bypassing them with identity connections

- *ResBlock update rule*:

$$H_\ell = \mathtt{ReLU}(b_\ell f_\ell(H_{\ell-1}) + \mathrm{id}(H_{\ell-1}))$$

  - b_ℓ ~ **Bernoulli(p_ℓ)**
- *Survival probabilities*:

$$p_\ell = 1 - \frac{\ell}{L}(1 - p_L), \;\; p_L = 0.5$$

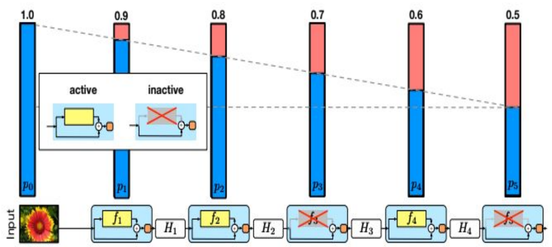

Fig. 1 The survival probability decays linearly, so deeper residual blocks are more likely to be skipped during training

Our Goal:

★ Train a 110-layer ResNet with *stochastic depth* on CIFAR-10 against *constant-depth* baseline

★ Verify test error and training time

★ Validate vanishing gradient alleviation, feature representation and generalizability

Methodology

1. Architecture

110-layer ResNet: 3 x 18 residual blocks, with 16, 32, and 64 filters respectively

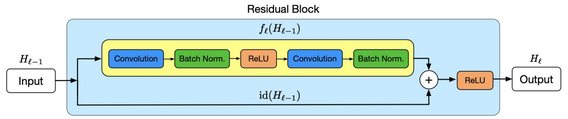

Fig. 2 Composition of a ResBlock in ResNet[1]

2. Training Setup

Setting	Configuration
Dataset	CIFAR-10 (45k train / 5k val / 10k test, 32×32 color images across 10 classes)
Data aug	horizontal flip + random 4px translation
Optimizer	SGD (momentum 0.9 + weight decay 1e-4)
lr schedule	lr=0.1 (10x decay at epochs 250, 375), 500 epochs total
Batch size	128
Framework	PyTorch (original in Torch 7)
Device	A100 GPU

3. Experiments

A. Core Reproduction of *training loss, final test error and wall-clock training time*

B. Gradient Magnitude Analysis

C. Feature Representation Quality

Metric	What it measures	Better =
t-SNE	Whether similar data cluster together in high-dimensional space	--
k-NN Accuracy	How well raw features classify unseen samples	↑ Higher
Silhouette Score	Cluster cohesion vs. separation in 64D	↑ Higher
Intra / Inter-class Ratio	Compactness within class vs. spread across classes	↓ Lower

D. Regularization Under Label Noise

Results

A. Stochastic Depth Reduces Test Error and Training Time

Method	Ours	Paper's
Constant depth	2h 51m	20h 42m
Stochastic depth	2h 28m	15h 7m
Wall-clock speedup	**13.2%**	~25%

8x time reduction!

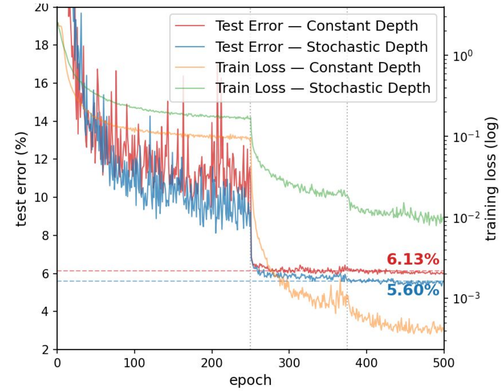

Fig. 3 Test Error and Training Loss on CIFAR-10

- Constant Depth:
  lower training loss → stronger overfitting
- Stochastic Depth:
  higher training loss → better generalization

B. Improved Gradient Strength

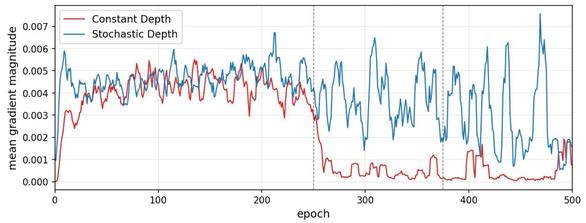

Fig. 4 Stochastic Depth Alleviates Vanishing Gradients

- Stochastic depth alleviates the vanishing gradient problem in very deep residual networks

C. Tighter Clusters ≠ Better Features

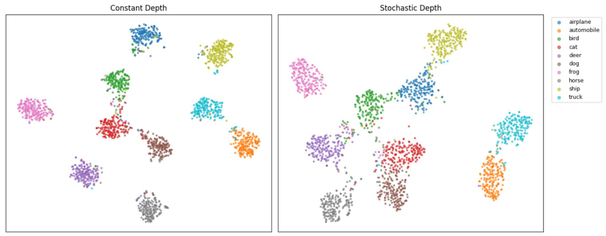

Fig. 5 t-SNE of penultimate layer features

Metric	Constant Depth	Stochastic Depth
k-NN Accuracy (k=1)	91.20%	91.40%
k-NN Accuracy (k=5)	93.00%	94.20%
k-NN Accuracy (k=10)	93.00%	94.60%
k-NN Accuracy (k=20)	93.00%	94.40%
Silhouette Score (higher = better)	0.6556	0.5377
Intra/Inter Distance Ratio (lower = better)	0.2524	0.3898

- Compactness reflects over-specialization, not richer representations

D. Slower Overfitting and More Robust Test Accuracy Under Label Noise

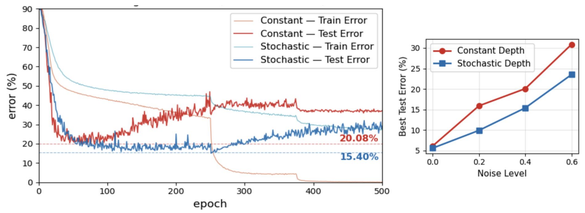

Fig. 6 Left: Testing and Training Error at Noise Level of 0.4
Right: Best Test Error Across Different Noise Levels

Conclusions

★ Stochastic depth: better accuracy and faster training on CIFAR-10

★ Shorter training paths preserve stronger gradients and ease deep ResNet optimization

★ Feature & label-noise analyses confirm stronger generalization against overfitting

References

[1] G. Huang, Y. Sun, Z. Liu, D. Sedra, and K. Q. Weinberger, "Deep networks with stochastic depth," *arXiv preprint arXiv:1603.09382*, 2016.